

Vectors

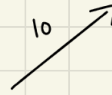
- a Vector has magnitude and direction

$$\vec{a} = \langle 2, 3, 4 \rangle$$

↑ the "→" symbolizes

a vector

↑ a vector has $\langle \rangle$ around it



↑ this is also a vector
it has direction and
magnitude (amount)

- Standard basic Vector

$$\vec{i} = \langle 1, 0, 0 \rangle$$

$$\vec{j} = \langle 0, 1, 0 \rangle$$

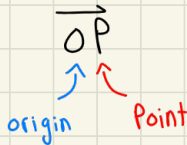
$$\vec{k} = \langle 0, 0, 1 \rangle$$

• these are called
Standard basic Vectors

you can think of these like they are x, y, z

you can write $\vec{a}$ like $\vec{a} = 2\vec{i} + 3\vec{j} + 4\vec{k}$

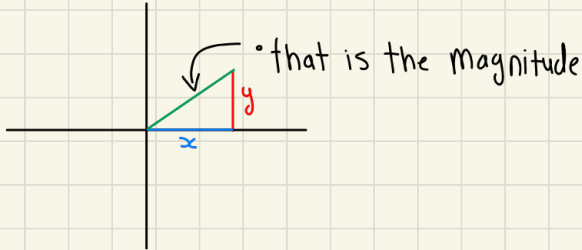
- a vector goes from the origin (0,0,0) to the Point (can be any point)
So we say:



- magnitude

- the distance between o origin and the point

$$|\vec{v}| = \sqrt{i^2 + j^2 + k^2}$$



the magnitude for $\vec{a} = \langle 2, 3, 4 \rangle$

$$|\vec{a}| = \sqrt{(2)^2 + (3)^2 + (4)^2} = \sqrt{29}$$

- unit vectors

- Unit Vector has magnitude 1 i, j and k are unit vectors

Find a vector that has the same direction as $\langle 7, -2, 3 \rangle$ and has unit length

$$|\vec{v}| = \sqrt{(7)^2 + (-2)^2 + (3)^2} = \sqrt{62}$$

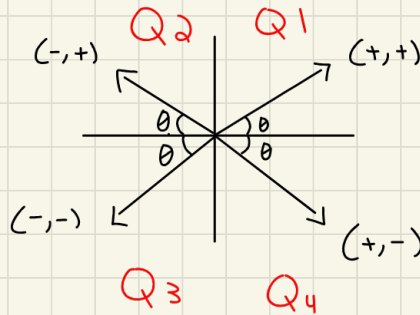
- once you have found the magnitude and divide it by the vector

$$\left(\frac{7}{\sqrt{62}}, \frac{-2}{\sqrt{62}}, \frac{3}{\sqrt{62}} \right)$$

- you can also write it like this

$$\frac{7}{\sqrt{62}}i - \frac{2}{\sqrt{62}}j + \frac{3}{\sqrt{62}}k$$

- breaking Vectors into their Components



- breaking a vector into its component in **Q1** will give you x-component that's positive and a y-component that's also positive
- breaking a vector in **Q2** you will get an x-component that's negative and a y-component that's positive
- breaking a vector in **Q3** you will get both the x and y component as negative
- breaking a vector in the **Q4** will result in a positive x-component and a negative y-component
- equation for breaking vectors into their components

$$x = |\vec{v}| \cos \theta$$

$$y = |\vec{v}| \sin \theta$$

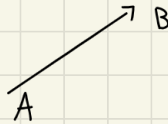
- Find the unit vector corresponding to $\vec{a}$

$$\vec{a} = \langle 4, 5 \rangle$$

$$|\vec{a}| = \sqrt{(4)^2 + (5)^2} = \sqrt{41}$$

$$\left(\frac{4}{\sqrt{41}}, \frac{5}{\sqrt{41}} \right)$$

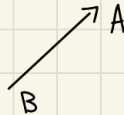
lets say we have this Vector at the head it's labeled "B" and on the tail its labeled "A"



We name it:

$$\overrightarrow{AB}$$

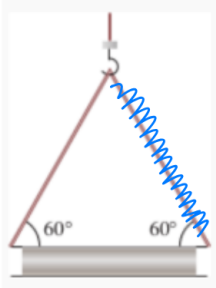
lets say we have this Vector at the head it's labeled "A" and on the tail its labeled "B"



We named it:

$$\overrightarrow{BA}$$

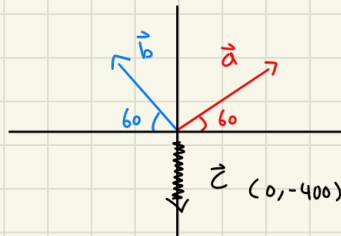
8. A crane suspends 400 lb beam horizontally by support cables, (with negligible weight) attached from a hook to each end of the beam. The support cable each makes an angle 60° with the beam. Find the tension vector in each support cable and magnitude of each tension.



C

- When you are dealing with Polygons (triangles) the Vectors will add up to Zero

B)



- you can move vectors around

$$a = \langle |\vec{a}| \cos(60^\circ), |\vec{a}| \sin(60^\circ) \rangle$$

$$b = \langle -|\vec{b}| \cos(60^\circ), |\vec{b}| \sin(60^\circ) \rangle$$

- the x-component for b is negative because it's in Q2

- We know that $\vec{a} + \vec{b} + \vec{c} = 0$

$$a = \langle |\vec{a}| \cos(60^\circ), |\vec{a}| \sin(60^\circ) \rangle$$

$$b = \langle -|\vec{b}| \cos(60^\circ), |\vec{b}| \sin(60^\circ) \rangle$$

$$c = \langle 0, -400 \rangle$$

the only thing we don't know is $|\vec{b}|$ and $|\vec{a}|$ but we do know what $\vec{c}$
 So we can do $\vec{a} + \vec{b} = -\vec{c}$

$$x\text{-values} \Rightarrow |\vec{a}| \cos(60^\circ) - |\vec{b}| \cos(60^\circ) = 0$$

$$|\vec{a}| \cancel{\cos(60^\circ)} = |\vec{b}| \cancel{\cos(60^\circ)}$$

$$|\vec{a}| = |\vec{b}|$$

• it equal to zero because
 the x-component in "c"
 is zero plus the whole
 thing is balance which
 the $|\vec{b}|$ and $|\vec{a}|$ are
 equal to "c" components

$$y\text{-values} \Rightarrow |\vec{a}| \sin(60^\circ) + |\vec{b}| \sin(60^\circ) = -(-400)$$

$$|\vec{b}| \sin(60^\circ) + |\vec{b}| \sin(60^\circ) = 400$$

$$|\vec{b}| (\sin(60^\circ) + \sin(60^\circ)) = 400$$

$$|\vec{b}| = \frac{400}{(\sin(60^\circ) + \sin(60^\circ))} = \frac{400}{\sqrt{3}}$$

$$a = \langle |\vec{a}| \cos(60^\circ), |\vec{a}| \sin(60^\circ) \rangle = \left(\frac{400}{\sqrt{3}} \cdot \frac{1}{2}, \frac{400}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \right)$$

$$b = \langle -|\vec{b}| \cos(60^\circ), |\vec{b}| \sin(60^\circ) \rangle = \left(-\frac{400}{\sqrt{3}} \cdot \frac{1}{2}, \frac{400}{\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \right)$$

$$\left(\frac{200}{\sqrt{3}}, 200 \right)$$

$$\left(-\frac{200}{\sqrt{3}}, 200 \right)$$

• Breaking Vectors into their Components Practice

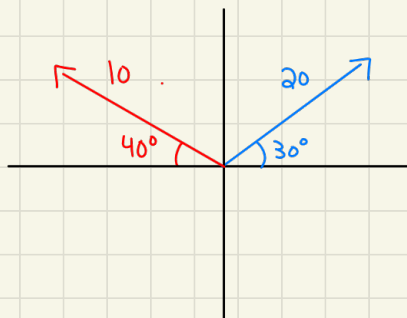
$$x = |\vec{V}| \cos \theta$$

$$x = 110 \cos(40) = -7.6$$

$$y = |\vec{V}| \sin \theta$$

$$y = 110 \sin(40)$$

$$y = 6.4$$



$$x = |\vec{V}| \cos \theta$$

$$x = 120 \cos(30) = 10\sqrt{3}$$

$$y = |\vec{V}| \sin \theta$$

$$y = 120 \sin(30) = 10$$

• the x is negative because its in the Q2

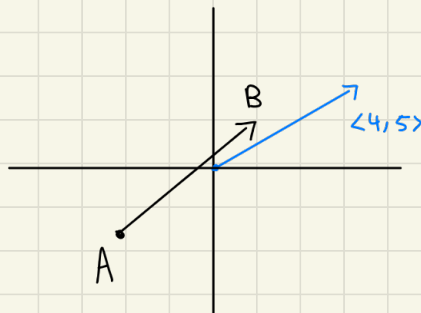
Given Vector $\vec{a}$ with represented by a differential line from $A(-3, -2)$ to $B(1, 3)$

- Find Vector $\vec{a}$

$$\vec{AB} = B - A \quad \text{Final - initial}$$

$$\langle (1 - (-3)), (3 - (-2)) \rangle$$

$$= \langle 4, 5 \rangle \quad \text{or you can write it like } 4i + 5j$$



$$\vec{a} = \langle 4, 5 \rangle$$

• always start from the origin when you want to plot